

TRIGONOMETRIC EQUATION

JEE EXERCISES

Exercise J-I

GENERAL SOLUTIONS OF TRIGONOMETRIC EQUATIONS

1. The most general solution of $\tan \theta = -1$, $\cos \theta = \frac{1}{\sqrt{2}}$ is :

- (A) $n\pi + \frac{7\pi}{4}$ (B) $n\pi + (-1)^n \frac{7\pi}{4}$ (C) $2n\pi + \frac{7\pi}{4}$ (D) None of these

निम्न का व्यापक हल करें $\tan \theta = -1$, $\cos \theta = \frac{1}{\sqrt{2}}$:

- (A) $n\pi + \frac{7\pi}{4}$ (B) $n\pi + (-1)^n \frac{7\pi}{4}$ (C) $2n\pi + \frac{7\pi}{4}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\tan \theta = -1 = \tan\left(-\frac{\pi}{4}\right); \quad \cos \theta = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4}$

$$\Rightarrow \theta = n\pi - \frac{\pi}{4}; \quad \theta = 2n\pi \pm \frac{\pi}{4}$$

$$\text{If } n=0 \text{ then } \theta = -\frac{\pi}{4}; \quad \theta = \frac{9\pi}{4}, \frac{7\pi}{4}$$

$$n=2 \text{ then } \theta = \frac{7\pi}{4}; \quad \theta = \frac{17\pi}{4}, \frac{15\pi}{4}$$

⋮

$$\Rightarrow \theta = \dots, -\frac{\pi}{4}, \frac{7\pi}{4}, \frac{15\pi}{4}, \dots$$

$$\Rightarrow \theta = 2n\pi + \frac{7\pi}{4}; n \in \mathbb{I}$$

SOLVING TRIGONOMETRIC EQUATION TYPE 1 (FACTORISATION)

2. $\frac{\sin 3\theta}{2 \cos 2\theta + 1} = \frac{1}{2}$, if:

यदि $\frac{\sin 3\theta}{2 \cos 2\theta + 1} = \frac{1}{2}$ तो θ का मान ज्ञात कीजिए :

- (A) $\theta = n\pi + \frac{\pi}{6}$ (B) $\theta = 2n\pi - \frac{\pi}{6}$ (C) $\theta = n\pi + (-1)^n \frac{\pi}{6}$ (D) $\theta = n\pi - \frac{\pi}{6}$

Ans. C

Sol. $\frac{\sin 3\theta}{2 \cos 2\theta + 1} = \frac{1}{2} \Rightarrow 2 \sin 3\theta = 2 \cos 2\theta + 1$
 $\Rightarrow 2(3 \sin \theta - 4 \sin^3 \theta) = 2(1 - 2 \sin^2 \theta) + 1$
 $\Rightarrow 6 \sin \theta - 8 \sin^3 \theta = 2 - 4 \sin^2 \theta + 1$
 $\Rightarrow 8 \sin^3 \theta - 4 \sin^2 \theta = 6 \sin \theta - 3$
 $\Rightarrow (2 \sin \theta - 1)(4 \sin^2 \theta - 3) = 0$
 $\Rightarrow \sin \theta = \frac{1}{2} \text{ or } \sin^2 \theta = \frac{3}{4} = \sin^2 \left(\frac{\pi}{3} \right)$
 $\Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{6} \text{ or } \theta = n\pi \pm \frac{\pi}{3}$

3. Find no. of roots of the equation $\tan x + \sec x = 2 \cos x$ in the interval $[0, 2\pi]$:

$[0, 2\pi]$ में समीकरण $\tan x + \sec x = 2 \cos x$ के मूलों की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\tan x + \sec x = 2 \cos x$; $x \in [0, 2\pi]$
 $\Rightarrow \sin x + 1 = 2 \cos^2 x$; $x \neq \frac{\pi}{2}, \frac{3\pi}{2}$
 $\Rightarrow \sin x = -1$ or $\sin x = \frac{1}{2}$
In $0 \leq x \leq 2\pi$
 $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$
& $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$
 $\Rightarrow$ No. of roots is 2 as $\tan x$ is not defined at $x = \frac{3\pi}{2}$

4. The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2 \sin^2 x + 5 \sin x - 3 = 0$ is :

अन्तराल $[0, 3\pi]$ में x के उन मानों की संख्या जो समीकरण $2 \sin^2 x + 5 \sin x - 3 = 0$ को सन्तुष्ट करते हैं :

(A) 6 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $2 \sin^2 x + 5 \sin x - 3 = 0$; $x \in [0, 3\pi]$
 $\Rightarrow \sin x = -3$ or $\sin x = \frac{1}{2}$

$$\because \sin x \in [-1, 1]$$

$$\Rightarrow \sin = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$

$\Rightarrow$ No. of values of x is 4.

SOLVING TRIGONOMETRIC EQUATION TYPE 2 (a sin x + b cos x = c)

5. If $0 < x < \pi$, and $\cos x + \sin x = \frac{1}{2}$, then tan x is :

यदि $0 < x < \pi$, तथा $\cos x + \sin x = \frac{1}{2}$, तो tan x का मान है :

- (A) $(4 - \sqrt{7})/3$ (B) $-(4 + \sqrt{7})/3$ (C) $(1 + \sqrt{7})/4$ (D) $(1 - \sqrt{7})/4$

Ans. B

Sol. $0 < x < \pi$ and $\cos x + \sin x = 1/2$

$$2(\cos x + \sin x) = 1$$

Square both sides

$$\Rightarrow 4[\cos^2 x + \sin^2 x + 2\sin x \cos x] = 1$$

$$\Rightarrow 4[1 + \tan^2 x + 2\tan x] = \sec^2 x$$

$$\Rightarrow 4[1 + \tan^2 x + 2\tan x] = 1 + \tan^2 x$$

$$\Rightarrow 4 + 4\tan^2 x + 8\tan x = 1 + \tan^2 x$$

$$\Rightarrow 3\tan^2 x + 8\tan x + 3 = 0$$

$$\tan x = \frac{-8 \pm \sqrt{64 - 4 \cdot 3 \cdot 3}}{2 \cdot 3} = \frac{-8 \pm 2\sqrt{7}}{2 \cdot 3} = \frac{-4 \pm \sqrt{7}}{3}$$

6. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then :

माना $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ तथा $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ दो समुच्चय है तो :

- (A) $P \subset Q$ and $Q - P \neq \phi$ (B) $Q \not\subset P$
(C) $P \not\subset Q$ (D) $P = Q$

Ans. D

Sol. $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$

$$\Rightarrow \tan \theta = \sqrt{2} + 1 \quad \dots(i)$$

$$Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 \quad \dots(ii)$$

from (i) & (ii)

$$\Rightarrow P = Q$$

SOLVING TRIGONOMETRIC EQUATION TYPE 3 (SUM TO PRODUCT AND PRODUCT TO SUM)

7. The general solution of, $\sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x$ is :

$\sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x$ का व्यापक हल है :

(A) $n\pi + \frac{\pi}{8}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$ (C) $(-1)^n \left(\frac{n\pi}{2}\right) + \frac{\pi}{8}$ (D) $2n\pi + \cos^{-1}\left(\frac{3}{2}\right)$

Ans. B

Sol. $\sin x - 3 \sin 2x + \sin 3x = \cos x - 3 \cos 2x + \cos 3x$
 $\Rightarrow 2 \sin 2x \cos x - 3 \sin 2x = 2 \cos 2x \cos x - 3 \cos 2x$
 $\Rightarrow (\sin 2x - \cos 2x)(2 \cos x - 3) = 0$

$\Rightarrow \sin 2x = \cos 2x$ or $\cos x = \frac{3}{2}$

$\Rightarrow \tan 2x = 1$ or $\cos x = \frac{3}{2} > 1$
 Not possible as $\cos x \in [-1, 1]$

$\Rightarrow 2x = n\pi + \frac{\pi}{4}$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in \mathbb{I}$

8. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are :

[AIEEE-2011]

$\in (0, \pi)$ के ऐसे संभव मान कि $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ हैं, हैं :

[AIEEE-2011]

(A) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$ (B) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(C) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$ (D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Ans. A

Sol $\sin \theta + \sin 4\theta + \sin 7\theta = 0$; $\theta \in (0, \pi)$

$2 \sin\left(\frac{\theta+7\theta}{2}\right) \cos\left(\frac{7\theta-\theta}{2}\right) + \sin 4\theta = 0$

$\Rightarrow \sin 4\theta [2 \cos 3\theta + 1] = 0$

$\Rightarrow \sin 4\theta = 0 \Rightarrow 4\theta = 0, \pi, 2\pi, 3\pi, 4\pi$

$\Rightarrow \theta = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$

but 0 and π are not included.

$$\text{and } 2\cos 3\theta + 1 = 0 \Rightarrow \cos 3\theta = \frac{-1}{2}$$

$$\Rightarrow 3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$$

$$\Rightarrow \theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{10\pi}{9}$$

$$\text{but } \frac{10\pi}{9} \notin (0, \pi)$$

$$\text{So, } \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

9. If $0 \leq x < 2\pi$, then the number of real values of x , which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is: **[JEE(Main)2016]**

यदि $0 \leq x < 2\pi$ है, तो x के उन वास्तविक मानों की संख्या जो समीकरण $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$ को सन्तुष्ट करते हैं, है: **[JEE(Main)2016]**

- (A) 7 (B) 9 (C) 3 (D) 5

Ans. A

Sol Given $x \in [0, 2\pi)$

$$\cos x + \cos 2x + \cos 3x + \cos 4x = 0$$

$$\Rightarrow 2\cos \frac{5x}{2} \left(\cos \frac{3x}{2} + \cos \frac{x}{2} \right) = 0$$

$$\Rightarrow 4\cos \frac{5x}{2} \cos x \cos \frac{x}{2} = 0$$

$$\Rightarrow \cos \frac{5x}{2} = 0 \text{ or } \cos x = 0 \text{ or } \cos \frac{x}{2} = 0$$

$$\text{If } \cos \frac{5x}{2} = 0 \text{ then}$$

$$x = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$$\text{if } \cos x = 0 \text{ then}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\text{if } \cos \frac{x}{2} = 0 \text{ then}$$

$$x = \pi$$

There are total real values of x is 7.

10. If sum of all the solution of the equation $8 \cos x \cdot \left(\cos \left(\frac{\pi}{6} + x \right) \cdot \cos \left(\frac{\pi}{6} - x \right) - \frac{1}{2} \right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to : [JEE(Main)2018]

यदि समीकरण $8 \cos x \cdot \left(\cos \left(\frac{\pi}{6} + x \right) \cdot \cos \left(\frac{\pi}{6} - x \right) - \frac{1}{2} \right) = 1$ के अंतराल $[0, \pi]$ में सभी हलों का योग $k\pi$ है, तो k बराबर है :

[JEE(Main)2018]

- (A) $\frac{8}{9}$ (B) $\frac{20}{9}$ (C) $\frac{2}{3}$ (D) $\frac{13}{9}$

Ans. D

Sol. $8 \cos x \cdot \cos \left(\frac{\pi}{6} + x \right) \cos \left(\frac{\pi}{6} - x \right) - 4 \cos x = 1$

$$\Rightarrow 8 \cos x \left(\frac{3}{4} - \sin^2 x \right) - 4 \cos x = 1$$

$$\Rightarrow 2 \cos x \left(\frac{4 \cos^2 x - 1}{4} \right) - 4 \cos x = 1$$

$$\Rightarrow 8 \cos^3 x - 2 \cos x - 4 \cos x = 1$$

$$\Rightarrow 8 \cos^3 x - 6 \cos x = 1$$

$$\Rightarrow \cos 3x = 1/2$$

$$\Rightarrow 3x = 2n\pi \pm \frac{\pi}{3}$$

$$\text{Given } x \in [0, \pi]$$

$$\Rightarrow x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$$\text{Sum} = \frac{13\pi}{9}$$

11. If $5(\tan^2 x - \cos^2 x) = 2 \cos 2x + 9$, then the value of $\cos 4x$ is :

[JEE(Main)2017]

यदि $5(\tan^2 x - \cos^2 x) = 2 \cos 2x + 9$ है, तो $\cos 4x$ का मान है :

[JEE(Main)2017]

- (A) $\frac{2}{9}$ (B) $-\frac{7}{9}$ (C) $-\frac{3}{5}$ (D) $\frac{1}{3}$

Ans. B

Sol. $5(\tan^2 x - \cos^2 x) = 2 \cos 2x + 9$

$$\Rightarrow 5 \tan^2 x - 5 \cos^2 x = 2 \cos 2x + 9$$

$$\begin{aligned}
&\Rightarrow 5\left(\frac{1-\cos 2x}{1+\cos 2x}\right) - 5\left(\frac{1+\cos 2x}{2}\right) = 2\cos 2x + 9 \\
&\Rightarrow 10(1-\cos 2x) - 5(1+\cos 2x)^2 = 2(1+\cos 2x)(2\cos 2x + 9) \\
&\Rightarrow 10 - 10\cos 2x - 5(1 + \cos^2 2x + 2\cos 2x) \\
&= 2[2\cos 2x + 9 + 2\cos^2 2x + 9\cos 2x] \\
&= 5 - 20\cos 2x - 5\cos^2 2x = 4\cos 2x + 18 + 18\cos 2x + 4\cos^2 2x \\
&\Rightarrow 9\cos^2 2x + 42\cos 2x + 13 = 0 \\
&\Rightarrow \left(\cos 2x + \frac{1}{3}\right)(9\cos 2x + 39) = 0 \\
&\Rightarrow \cos 2x = -\frac{1}{3} \quad \text{or} \quad \cos 2x = \frac{-39}{9} = \frac{-13}{3} < -1 \\
&\Rightarrow \cos 2x = -\frac{1}{3} \\
&\Rightarrow \cos 4x = 2\cos^2 2x - 1 = \frac{2}{9} - 1 = -\frac{7}{9}
\end{aligned}$$

SOLVING TRIGONOMETRIC EQUATION TYPE 5 (MAXIMUM AND MINIMUM BASED)

12. If $\theta \in [0, 5\pi]$ and $r \in \mathbb{R}$ such that $2\sin \theta = r^4 - 2r^2 + 3$ then the maximum number of values of the pair (r, θ) is :
 (A) 8 (B) 10 (C) 6 (D) none of these

यदि $\theta \in [0, 5\pi]$ और $r \in \mathbb{R}$ इस प्रकार है कि $2\sin \theta = r^4 - 2r^2 + 3$ तब हल (r, θ) कि संख्या ज्ञात करें।

- (A) 8 (B) 10 (C) 6 (D) इनमें से कोई नहीं

Ans. C

Sol. $\theta \in [0, 5\pi]; r \in \mathbb{R}$

$$2\sin \theta = r^4 - 2r^2 + 3$$

$$\Rightarrow 2\sin \theta = (r^2 - 1)^2 + 2$$

$$\therefore 2\sin \theta \in [-2, 2] \text{ \& } (r^2 - 1)^2 + 2 \in [2, \infty)$$

$\Rightarrow$ It is possible if

$$2\sin \theta = 2 \text{ \& } (r^2 - 1)^2 + 2 = 2$$

$$\Rightarrow \sin \theta = 1 \text{ \& } (r^2 - 1)^2 = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} \text{ \& } r = \pm 1$$

$\Rightarrow$ Maximum number of values of the pair (r, θ) is 6.

13. The number of real solutions of, $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$ is :

- (A) Zero (B) One (C) Two (D) Infinite

समीकरण $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$ के वास्तविक हलों की संख्या होगी :

- (A) शून्य (B) एक (C) दो (D) अन्त

Ans. A

Sol. $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$

$$\Rightarrow \sin e^x \cdot \cos e^x = \frac{2^x}{4} + \frac{2^{-x}}{4}$$

$$\Rightarrow 4 \sin e^x + \cos e^x = 2^x + 2^{-x}$$

$$\Rightarrow 2 \sin (2e^x) = 2^x + 2^{-x}$$

$$\therefore 2 \sin (2e^x) \in [-2, 2]$$

$$\& 2^x + 2^{-x} \in [2, \infty)$$

$\Rightarrow$ This is possible if

$$2 \sin (2e^x) = 2 \& 2^x + 2^{-x} = 2$$

$$\Rightarrow \underbrace{\sin(2e^x) = 1}_{\text{This is not possible if } x=0} \& x = 0$$

$\Rightarrow$ No. of solution is zero.

14. The general solution of the equation, $\cos x \cos 6x = -1$ is :

- (A) $x = (2n + 1)\pi, n \in I$ (B) $x = 2n\pi, n \in I$
(C) $x = (4n - 1)\pi, n \in I$ (D) None

समीकरण $\cos x \cos 6x = -1$ का व्यापक हल होगा :

- (A) $x = (2n + 1)\pi, n \in I$ (B) $x = 2n\pi, n \in I$
(C) $x = (4n - 1)\pi, n \in I$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\cos x \cos 6x = -1$

$$\Rightarrow 2 \cos x \cos 6x = -2$$

$$\Rightarrow \cos 7x + \cos 5x = -2$$

$$\therefore \cos x \in [-1, 1]$$

$$\Rightarrow \cos 7x = -1 \& \cos 5x = -1$$

$$\Rightarrow 7x = (2n + 1)\pi; n \in I \& 5x = (2n + 1)\pi; n \in I$$

$$\Rightarrow x = \left(\frac{2n+1}{7}\right)\pi; n \in I \& x = \left(\frac{2n+1}{5}\right)\pi; n \in I$$

$$\Rightarrow x = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

$$\Rightarrow x = (2n + 1)\pi; n \in I$$

15. The number of solutions of the equation, $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$ is :

- (A) Zero (B) Only one (C) Only two (D) Greater than 2

समीकरण $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$ के हलों की संख्या है :

- (A) शून्य (B) केवल एक (C) केवल दो (D) दो से बड़ा

Ans. B

Sol. $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$

$$\because x^2 - 2\sqrt{3}x + 4 \in [1, \infty)$$

$$\& \sin\left(\frac{\pi x}{2\sqrt{3}}\right) \in [-1, 1]$$

$\Rightarrow$ This is possible if

$$\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = 1 \& x^2 - 2\sqrt{3}x + 4 = 1$$

$$\Rightarrow \frac{\pi x}{2\sqrt{3}} = (4n+1)\frac{\pi}{2}; n \in I \& (x - \sqrt{3})^2 = 0$$

$$x = \sqrt{3}(4n+1); n \in I \& x = \sqrt{3}$$

$\Rightarrow$ No. of solution is 1.

16. General solution for θ if, $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$ is :

- (A) $2n\pi + \frac{7\pi}{6}$ (B) $2n\pi + \frac{\pi}{6}$ (C) $2n\pi - \frac{7\pi}{6}$ (D) None of these

समीकरण $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$ के लिए θ का व्यापक हल है :

- (A) $2n\pi + \frac{7\pi}{6}$ (B) $2n\pi + \frac{\pi}{6}$ (C) $2n\pi - \frac{7\pi}{6}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$

$$\because \sin\left(2\theta + \frac{\pi}{6}\right) \in [-1, 1]$$

$$\& \cos\left(\theta + \frac{5\pi}{6}\right) \in [-1, 1]$$

This is only possible if

$$\sin\left(2\theta + \frac{\pi}{6}\right) = 1 \text{ \& } \cos\left(\theta + \frac{5\pi}{6}\right) = 1$$

$$\Rightarrow 2\theta + \frac{\pi}{6} = (4n+1)\frac{\pi}{2}; n \in I \text{ \& } \theta + \frac{5\pi}{6} = 2n\pi; n \in I$$

$$\Rightarrow 2\theta = 2n\pi + \frac{\pi}{3}; n \in I \text{ \& } \theta = 2n\pi - \frac{5\pi}{6}; n \in I$$

$$\Rightarrow \theta = n\pi + \frac{\pi}{6}; n \in I \text{ \& } \theta = 2n\pi - \frac{5\pi}{6}; n \in I$$

$\Rightarrow$ General solution for θ is

$$2n\pi + \frac{\pi}{6}; n \in I$$

17. The number of values of 'x' for which $\sin 2x + \cos 4x = 2$ is :

- (A) 0 (B) 1 (C) 2 (D) infinite

'x' के मानों की संख्या ज्ञात करें जो समीकरण $\sin 2x + \cos 4x = 2$ को संतुष्ट करते हैं :

- (A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. A

Sol. $\sin 2x + \cos 4x = 2$

$$\because \sin 2x \in [-1, 1] \text{ \& } \cos 4x \in [-1, 1]$$

$\Rightarrow$ This is only possible if

$$\sin 2x = 1 \text{ \& } \cos 4x = 1$$

$$\Rightarrow 2x = (4n+1)\frac{\pi}{2}; n \in I \text{ \& } 4x = 2n\pi; n \in I$$

$$\Rightarrow x = n\pi + \frac{\pi}{4}; n \in I \text{ \& } x = \frac{n\pi}{2}; n \in I$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \dots \text{ \& } x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots$$

$\Rightarrow$ No. of value of x is zero.

18. The equation, $4 \sin^2 x + 4 \sin x + a^2 - 3 = 0$ possesses a solution if 'a' belongs to the interval :

यदि समीकरण $4 \sin^2 x + 4 \sin x + a^2 - 3 = 0$ का हल संभव होगा यदि 'a' निम्न अन्तराल में होगा :

- (A) $(-1, 3)$ (B) $(-3, 1)$ (C) $[-2, 2]$ (D) $R - (-2, 2)$

Ans. C

Sol. $4 \sin^2 x + 4 \sin x + a^2 - 3 = 0$

$$\Rightarrow 4 \sin^2 x + 4 \sin x = 3 - a^2$$

$$\Rightarrow 4[\sin^2 x + \sin x] = 3 - a^2$$

$$\Rightarrow \left[\left(\sin x + \frac{1}{2} \right)^2 \right] = 4 - a^2$$

$$\therefore \left(\sin x + \frac{1}{2} \right)^2 \in [0, 1]$$

$$\Rightarrow 4 \left(\sin x + \frac{1}{2} \right)^2 \in [0, 4]$$

$$\Rightarrow 4 - a^2 \in [0, 4]$$

$$\Rightarrow 0 \leq 4 - a^2 \leq 4$$

$$\Rightarrow -4 \leq -a^2 \leq 0$$

$$\Rightarrow 0 \leq a^2 \leq 4$$

$$\Rightarrow a \in [-2, 2]$$

19. The equation, $\cos 2x + a \sin x = 2a - 7$ possesses a solution if :

(A) $a < 2$

(B) $2 \leq a \leq +6$

(C) $a > 6$

(D) a is any integer

समीकरण $\cos 2x + a \sin x = 2a - 7$ का हल विद्यमान है, यदि :

(A) $a < 2$

(B) $2 \leq a \leq +6$

(C) $a > 6$

(D) a कोई भी पूर्णांक है।

Ans. B

Sol. $\cos 2x + a \sin x = 2a - 7$

i.e. $2\sin^2 x - a \sin x + 2a - 8 = 0$

$$\sin x = \frac{a \pm \sqrt{a^2 - 8(2a - 8)}}{4} = \frac{a \pm (a - 8)}{4}$$

$$\sin x = \frac{a - 4}{2} \text{ or } \underbrace{\sin x = 2}_{\text{not possible}}$$

Hence $-1 \leq (a - 4)/2 \leq 1 \Rightarrow a \in [2, 6]$

TRIGONOMETRIC INEQUALITIES

20. If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is :

यदि $0 < \theta < 2\pi$ तो θ का अन्तराल जिनके लिये $2\sin^2\theta - 5\sin\theta + 2 > 0$ है :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

(B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$

(C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

(D) $\left(\frac{41\pi}{48}, \pi\right)$

Ans. A

Sol. $2\sin^2\theta - 5\sin\theta + 2 > 0$

Given $\theta \in (0, 2\pi)$

$$\Rightarrow 2\sin^2\theta - 4\sin\theta - \sin\theta + 2 > 0$$

$$\Rightarrow (\sin\theta - 2)(2\sin\theta - 1) > 0$$

$$\therefore \sin\theta \in [-1, 1]$$

$$\Rightarrow \sin\theta - 2 < 0$$

$$\Rightarrow 2\sin\theta - 1 < 0$$

$$\Rightarrow \sin \theta < \frac{1}{2}$$

$$\because \theta \in (0, 2\pi)$$

$$\Rightarrow \theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$$

SYSTEM OF TRIGONOMETRIC EQUATION

21. $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = 1/e$, where $\alpha, \beta \in [-\pi, \pi]$, numbers of pairs of α, β which satisfy both the equations is :

$\cos(\alpha - \beta) = 1$ तथा $\cos(\alpha + \beta) = 1/e$, जहाँ $\alpha, \beta \in [-\pi, \pi]$, ऐसे कितने α, β के युग्म होंगे जो दोनों समीकरणों को संतुष्ट होंगे :

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $\cos(\alpha - \beta) = 1$ & $\cos(\alpha + \beta) = \frac{1}{e}$

Given $\alpha \in [-\pi, \pi]; \beta \in [-\pi, \pi]$

$\Rightarrow \alpha + \beta \in [-2\pi, 2\pi]; \alpha - \beta \in [-2\pi, 2\pi]$

$\because \cos(\alpha - \beta) = 1$

$\Rightarrow \alpha - \beta = -2\pi, 0, 2\pi$

$\Rightarrow (\alpha, \beta) = (-\pi, \pi), (\alpha, \alpha), (\pi, -\pi)$

For $\cos(\alpha + \beta) = \frac{1}{e}$; rejected value of $(\alpha, \beta) = (-\pi, \pi), (\pi, -\pi)$

$\Rightarrow \cos 2\alpha = \frac{1}{e}$

$\because 2\alpha \in [-2\pi, 2\pi] \text{ \& } \frac{1}{e} < 1$

$\Rightarrow$ No. of pairs of α, β are 4.

22. The number of solutions of the pair of equations $2 \sin^2 \theta - \cos 2\theta = 0$ $2 \cos^2 \theta - 3 \sin \theta = 0$ in the interval $[0, 2\pi]$ is: [JEE-2007]

- (A) zero (B) one (C) two (D) four

समीकरणों $2 \sin^2 \theta - \cos 2\theta = 0$, $2 \cos^2 \theta - 3 \sin \theta = 0$ के अन्तराल $[0, 2\pi]$ में हलों की संख्या है : [JEE-2007]

- (A) शून्य (B) एक (C) दो (D) चार

Ans. C

Sol. $2 \sin^2 \theta - \cos 2\theta = 0$, $2 \cos^2 \theta - 3 \sin \theta = 0$

$\Rightarrow 2 \sin^2 \theta - (1 - 2 \sin^2 \theta) = 0$; $2(1 - \sin^2 \theta) - 3 \sin \theta = 0$

$$\Rightarrow \sin^2 \theta = \frac{1}{4} \quad ; \quad 2\sin^2 \theta + 3\sin \theta - 2 = 0$$

$$\Rightarrow \sin^2 \theta = \frac{1}{4} \quad ; \quad \sin \theta = \frac{1}{2}$$

$\Rightarrow$ Number of solution of the pair of equation $2\sin^2 \theta - \cos 2\theta = 0$, $2\cos^2 \theta - 3\sin \theta = 0$ in the interval $[0, 2\pi]$ is 2.